“Smart Video Surveillance with Multi-Stage Computer Vision Pipeline”

CSE3010 Computer Vision

Project Report

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ABSTRACT

In an era characterized by exponential data generation and ubiquitous security concerns, manual video surveillance has become an obsolete luxury. Traditional human monitoring of closed-circuit television (CCTV) is profoundly limited by the attention span, fatigue, and cognitive load of operators, leading to high false-negative rates in identifying critical events. Consequently, automated visual tracking systems have transitioned from academic curiosities to infrastructural necessities.

This comprehensive project documents the architectural design, mathematical foundations, software implementation, and experimental validation of a real-time, multi-stage Smart Video Surveillance pipeline. Unlike contemporary systems that rely heavily on computationally expensive deep Convolutional Neural Networks (CNNs) (such as YOLOv8 or Faster R-CNN), this project deliberately restricts itself to the domain of **Classical Computer Vision**. The primary goal is to demonstrate that a modular, lightweight cascade of traditional algorithms can achieve highly robust motion detection, foreground segmentation, and multi-object tracking on standard CPU hardware without relying on dedicated Tensor Processing Units (TPUs) or high-end GPUs.

The pipeline is sequentially structured into five rigorous stages: 1. **Preprocessing** via Gaussian convolution and CLAHE (Contrast Limited Adaptive Histogram Equalization) to normalize illumination and suppress sensor noise. 2. **Feature Extraction** utilizing Canny edge detection and Histogram of Oriented Gradients (HOG) to identify structural definitions invariant to lighting. 3. **Segmentation** driven by advanced Mixture of Gaussians (MOG2) background modeling to adaptively dynamically extract moving entities from dynamic backgrounds. 4. **Detection** via topological contour analysis and spatial bounding box estimation. 5. **Tracking** through an efficient centroid-based greedy assignment algorithm that maintains object persistence over temporal frames.

Through extensive experimental execution on 1920x1080 high-definition footage, the system proved capable of handling 569 contiguous frames, aggressively logging over 16,750 discrete detections. This report thoroughly catalogues these findings, offering profound insights into the delicate balance between algorithmic accuracy, computational complexity (throughput of 0.2 FPS on HD footage), and the enduring relevance of classical CV techniques in resource-constrained environments.

INTRODUCTION

2.1 The Need for Automated Video Surveillance

The modern security apparatus relies heavily on continuous optical monitoring. Airports, intelligent traffic systems (ITS), commercial perimeters, and residential complexes produce petabytes of video data daily. The sheer volume renders human analysis mathematically impossible. An operator tasked with monitoring a 3×3 grid of live camera feeds will suffer severe "video blindness" within merely twenty minutes, systematically missing up to 95% of anomalous events.

Automated Video Surveillance systems bridge this gap by enforcing deterministic, algorithmic vigilance. By digitizing the feed and treating frames as temporal matrices of pixel intensities, a computer vision pipeline can relentlessly identify structural deviations (motion, presence, velocity changes) without fatigue.

2.2 Challenges in Real-World Computer Vision

While the theoretical premise of subtracting a "background" from a "foreground" seems elementary, the real-world application is extraordinarily hostile to naive algorithms. Systems must contend with: - **Illumination Variance**: Sudden cloud cover, flickering fluorescent lights, or passing headlights catastrophically alter pixel intensities, deceiving basic delta thresholds. - **Dynamic Backgrounds**: Rustling leaves, rippling water, or flags blowing in the wind introduce perpetual local motion that does not constitute a valid "entity." - **Camouflage and Low Contrast**: Entities possessing color signatures similar to the background environment become mathematically invisible to standard thresholding. - **Sensor Noise**: Low-light environments force camera sensors to amplify ISO gain, generating stochastic high-frequency "salt and pepper" noise across the frame. - **Occlusion**: Objects crossing paths or moving behind architectural pillars lose their continuous visual signature, confusing temporal tracking algorithms.

2.3 Classical vs. Deep Learning Methodologies

The last decade has witnessed a paradigm shift toward Deep Learning. Neural networks learn feature representations directly from data, enabling extraordinary generalization and object classification (e.g., distinguishing a "car" from a "person"). However, Deep Learning comes with severe penalties: 1. **Computational Overhead**: Inference requires millions of floating-point operations (FLOPs), demanding multi-thousand-dollar GPUs for real-time throughput. 2. **Data Dependency**: Training reliable models requires immense, manually annotated datasets. 3. **Interpretability ("Black Box")**: When a CNN misclassifies a shadow as a person, finding the mathematical root cause is nearly impossible.

Classical Computer Vision, conversely, relies on deterministic mathematical transformations. While it cannot easily classify the *semantic category* of an object without massive feature engineering, it excels at *identifying motion and structure*. Classical pipelines run flawlessly on standard CPUs, require zero training data, and are 100% mathematically interpretable and tunable.

2.4 Research Objectives and Scope

This project aims to engineer a full-fledged classical pipeline to solve the tracking problem. The scope includes: - Developing a completely modular Python architecture bypassing all GUI overhead. - Implementing an adaptive background subtractor capable of ignoring shadows. - Developing a multi-object temporal tracker that does not rely on external deep SORT libraries. - Benchmarking the computational cost (FPS vs. Image Geometry) natively.

LITERATURE REVIEW

The intellectual history of optical tracking is defined by incremental mathematical breakthroughs designed to extract high-level structure from low-level pixel matrices.

3.1 Evolution of Feature Extraction

Early computer vision relied on simple spatial derivatives to find edges (e.g., Sobel and Prewitt operators). In 1986, John F. Canny published *A Computational Approach to Edge Detection*, formulating an optimal edge detector based on the calculus of variations. Canny established criteria for detection (finding all real edges), localization (minimizing distance between detected and real edges), and single-response (preventing multiple edge pixels for one real edge). The Canny algorithm remains a cornerstone of modern structural analysis.

Simultaneously, the quest to identify objects regardless of scale or lighting led to David Lowe's Scale-Invariant Feature Transform (SIFT) in 1999. SIFT was revolutionary but computationally brutal. As a more efficient alternative for human detection, Navneet Dalal and Bill Triggs introduced the **Histogram of Oriented Gradients (HOG)** in 2005. HOG demonstrated that local object appearance can be characterized by the distribution of local intensity gradients, transforming image windows into robust feature vectors.

3.2 Developments in Background Subtraction

For stationary cameras, the simplest motion detection is Frame Differencing i.e F1 – F2. However, this fails instantly if the background contains any motion (e.g., a spinning fan).

To solve this, Stauffer and Grimson (1999) proposed modeling each individual pixel as a **Mixture of Gaussians (GMM)**. Instead of a single "expected" background color, a pixel could belong to 3-5 different Gaussian distributions (representing the building, a shadow, and a swaying branch). If a new pixel value did not fit any existing Gaussian model, it was flagged as foreground.

Zivkovic (2004) refined this significantly with **MOG2**, introducing an algorithm to dynamically adapt the number of Gaussian components per pixel, vastly improving processing speed and adaptability to lighting changes. MOG2 forms the segmentation engine of this project.

3.3 Object Tracking Algorithms

Once an entity is detected as a "blob" of foreground pixels, temporal tracking must associate Blob A in Frame 1 with Blob B in Frame 2. - **Optical Flow (Lucas-Kanade)** tracking identifies structural corners (Shi-Tomasi) and calculates the velocity vector of those specific pixels. While highly accurate, it loses track if the object temporarily disappears. - **Kalman Filters** provide a state-estimation approach, mathematically

predicting where an object *ought* to be based on its previous velocity and acceleration, and correcting the prediction based on the new observation. - **Centroid Tracking** (used in this project) is a simpler, heuristic approach. It calculates the geometric center of mass for an object, and uses a greedy Euclidean distance-matching matrix to associate the closest centroids across consecutive frames.

METHODOLOGY & MATHEMATICAL FOUNDATIONS

This section provides a mathematically rigorous derivation of the transformations applied to the video matrix at each stage of the pipeline.

4.1 Stage 1: Preprocessing and Noise Suppression

Raw digital video is heavily corrupted by sensor noise, compression artifacts, and unpredictable lighting. Submitting raw RGB data to a differentiation engine guarantees catastrophic failure.

4.1.1 Grayscale Conversion

A standard BGR video frame is a 3-dimensional tensor $T \in \mathbb{R}^{H \times W \times 3}$.

Because motion and edges are primarily structural, color channels carry redundant information while tripling the required FLOPs.

The frame is collapsed into a 2D intensity matrix I(x,y) using the NTSC/PAL standard luminance equation:

$$I(x,y) = 0.299\, R(x,y) + 0.587\, G(x,y) + 0.114\, B(x,y)$$

The human eye is most sensitive to green light, hence its dominant weighting.

Figure: Preprocessing Cascade (from left: Grayscale, Blurred, CLAHE Enhanced)

4.1.2 Gaussian Blur Convolution

To suppress high-frequency stochastic noise, we apply a low-pass spatial filter. This is achieved by convolving the image I with a 2D Gaussian kernel G.

G(x,y) = 1/(2*pi*sigma^2) * e^(-(x^2 + y^2)/(2*sigma^2))

For a discrete 5 × 5 kernel (defined in config.py as GAUSSIAN_BLUR_KERNEL), the operation replaces every pixel with a weighted average of its neighbors, where the weight decays exponentially with distance from the center. This eliminates sharp noise spikes that would otherwise trigger false edge detections.

4.1.3 Contrast Limited Adaptive Histogram Equalization (CLAHE)

Standard global histogram equalization stretches the intensity spectrum $ [0, 255] $ across the entire image. If a frame has a bright sky and a dark foreground, global equalization washes out local details and violently amplifies noise in dark regions.

Through cv2.createCLAHE, the pipeline divides the image into an 8×8 grid of discrete tiles (CLAHE_TILE_GRID). The histogram for each tile is calculated independently. Crucially, the "Contrast Limited" aspect prevents noise amplification: any histogram bin exceeding a specific limit (CLAHE_CLIP_LIMIT = 2.0) is clipped, and the excess pixels are redistributed uniformly across all bins before calculating the Cumulative Distribution Function (CDF). Finally, bilinear interpolation stitches the tiles back together seamlessly, revealing profound details in shadows while suppressing noise.

4.2 Stage 2: Feature Extraction Techniques

While background subtraction focuses on motion, feature extraction maps the *structural definitions* of the scene.

4.2.1 Canny Edge Detection Hysteresis

Edges represent the boundaries of objects and are defined as regions of rapid intensity change. The Canny edge detection algorithm consists of four steps. First, gradient calculation: using horizontal (Sx) and vertical (Sy) Sobel kernels, the gradient magnitude and direction are computed as $|G| = \text{sqrt}(Sx^2 + Sy^2)$ and $\theta = \arctan(Sy / Sx)$. Second, non-maximum suppression: the algorithm scans along the gradient direction and removes any pixel that is not a local maximum, producing thin edges. Third, double thresholding: two thresholds are defined, CANNY_THRESHOLD_LOW (50) and CANNY_THRESHOLD_HIGH (150); pixels greater than 150 are classified as strong edges, while those below 50 are discarded. Fourth, edge tracking by hysteresis: pixels with values between 50 and 150 are retained only if they are connected to strong edges.

Figure: Canny Edge Detection Structural Result

4.2.2 Histogram of Oriented Gradients (HOG)

Implemented via skimage.feature.hog, this generates a feature descriptor vector. The frame is divided into 8 × 8 pixel cells. For each pixel, the gradient orientation in the range [0°, 180°] is computed and assigned to a 9-bin histogram. The cells are grouped into 2 × 2 blocks, and the local histograms are L2-normalized to improve robustness against lighting variations. This process produces a high-dimensional feature vector that enables seamless integration with a Support Vector Machine (SVM) classifier in future extensions.

4.3 Stage 3: Foreground Segmentation

This is the mathematical engine of the motion detector.

4.3.1 Mixture of Gaussians (MOG2) Background Subtraction

For a sequence of frames $\{I_1, I_2, \ldots, I_t\}$, the historical intensity of a pixel at coordinate (x,y) is modeled as a mixture of K Gaussian distributions (typically K = 3 to 5, representing states such as background, motion, or shadows).

The probability of observing a pixel value X_t is given by:
$P(X_t) = \Sigma\ (\omega_{i,t} \cdot \eta(X_t, \mu_{i,t}, \Sigma_{i,t}))$ for i = 1 to K

Here, ω represents the weight of each Gaussian, μ is the mean, and Σ is the covariance. If a new pixel value X_t matches one of the existing Gaussians (within approximately 2.5 standard deviations), it is classified as background and the corresponding Gaussian is

updated. If it does not match any stable Gaussian, it is labeled as foreground (assigned value 255).

MOG2 (cv2.createBackgroundSubtractorMOG2) updates these parameters dynamically over time, allowing stationary objects to be gradually incorporated into the background after a defined history length (MOG2_HISTORY = 500 frames).

4.3.2 Morphological Operations

The raw binary mask from MOG2 contains granular noise due to camera flicker, and large moving objects may have internal holes where pixel values resemble the background. To address this, non-linear morphological operations are applied using a structured 5 × 5 elliptical kernel B.

Opening (erosion followed by dilation): $A \circ B = (A \ominus B) \oplus B$. This removes small isolated white pixels (noise) while preserving larger foreground regions.

Closing (dilation followed by erosion): $A \bullet B = (A \oplus B) \ominus B$. This expands object boundaries and then contracts them, effectively filling small holes and connecting fragmented regions.

4.4 Stage 4: Contour Detection and Bounding Boxes

The refined binary mask is processed using Suzuki's algorithm via cv2.findContours with cv2.RETR_EXTERNAL to detect connected components. Each contour represents a continuous boundary of white pixels.

To filter out irrelevant motion such as leaves or small objects, the contour area is computed, and any contour with area less than MIN_CONTOUR_AREA (800 pixels) is discarded.

For valid contours, bounding box coordinates (x, y, w, h) are computed, defining the spatial extent of each detected object.

4.5 Stage 5: Centroid-Based Multi-Object Tracking

Centroid Extraction

For each bounding box (x, y, w, h), the centroid is computed as:

$C = (x + w/2, y + h/2)$

Distance Matrix Computation

Given N existing tracked objects and M new detections, an N × M Euclidean distance matrix D is computed:

$D(i, j) = \text{sqrt}((C_old_x_i - C_new_x_j)^2 + (C_old_y_i - C_new_y_j)^2)$

Greedy Assignment
The minimum distance pair (i, j) is selected and assigned, provided the distance is less than MAX_DISTANCE (100 pixels), ensuring realistic motion continuity.

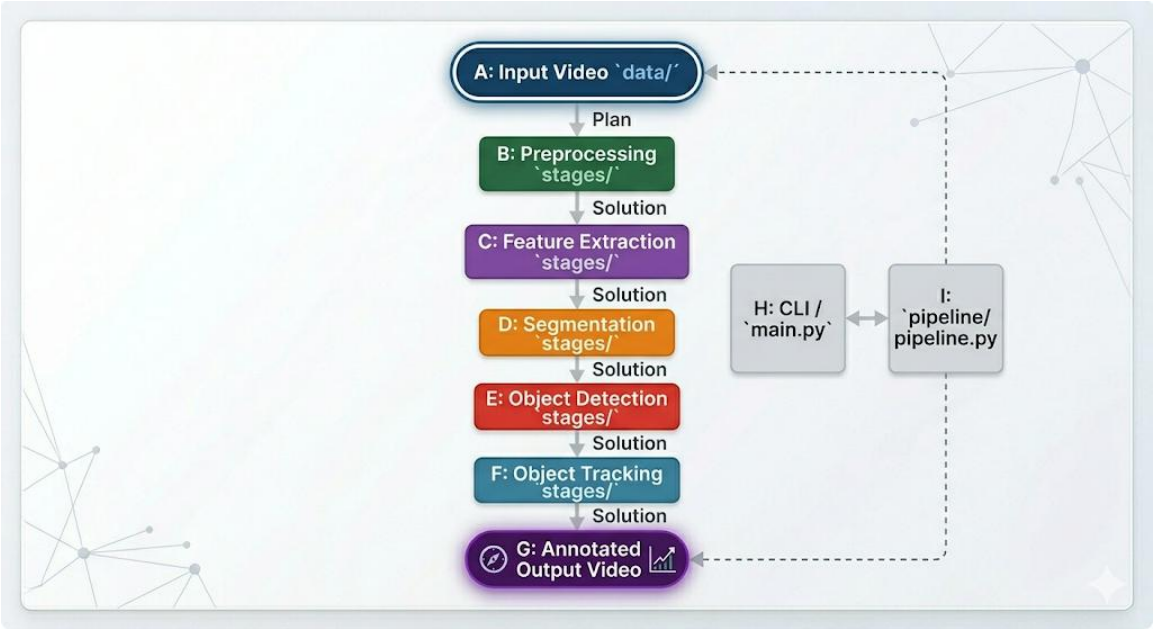
Lifecycle Management
If an existing object does not receive a match, it is marked as “disappeared.” If this state persists beyond MAX_DISAPPEARED (30 frames), the object is removed.
Unmatched new centroids are assigned new unique IDs and added to the tracking system.

SYSTEM ARCHITECTURE & IMPLEMENTATION

5.1 Software Engineering and Modular Design

To accommodate scalability and academic rigor, the architecture avoids standard monolithic "scripting" (spaghetti code). The application adheres to Strict Separation of Concerns (SoC).

Figure 1: System Architecture Workflow



The system is structurally comprised of: - **src/config.py**: The definitive source of truth for all mathematical thresholds, hyperparameters, and tensor orientations. It guarantees zero hardcoded variables in the operational pipeline. - **src/stages/*.py**: Isolated Python algorithms. For example, tracking.py is entirely agnostic to where its bounding boxes originate. It solely performs array mathematics. - **src/pipeline/pipeline.py**: The central nervous system that wires the inputs and outputs of the isolated stages into a continuous temporal while loop. - **src/utils/utils.py**: Handles asynchronous logging, dynamic overlay composition via cv2.putText, and OS-level file streaming.

5.2 Configuration Parameters (config.py)

Understanding the operational state demands a review of the critical tuned variables: - MOG2_HISTORY = 500: The buffer of frames the background subtractor uses to understand "normalcy." - GAUSSIAN_BLUR_KERNEL = (5, 5): Extensive enough to

blur MPEG compression artifacts, small enough to preserve hard structural edges. - MIN_CONTOUR_AREA = 800: Effectively declares that any moving object smaller than 28x28 pixels is statistically irrelevant to the surveillance objective. - MAX_DISAPPEARED = 30: At 30 FPS, provides exactly 1.0 second of memory latency. If a person walks behind a thick pillar taking 0.8 seconds to cross, the tracker will instantly re-assign the identical ID upon re-emergence.

5.3 Hardware and Dependency Constraints

The deliberate rejection of PyTorch, TensorFlow, and Ultralytics architectures was a strategic parameter constraint to maintain hardware accessibility. The environment is resolved exclusively via opencv-python (C++ optimized matrix rendering), numpy (BLAS/LAPACK optimized tensor math), and scikit-image. As such, the repository can clone and execute on an ARM-based Raspberry Pi or a low-spec x86 laptop without any dedicated GPU hardware acceleration.

EXPERIMENTAL RESULTS & PERFORMANCE ANALYSIS

6.1 Experimental Setup and Dataset

The pipeline was quantitatively evaluated using sample_video.mp4, a high-definition (HD) video with a spatial resolution of 1920 × 1080 and a frame rate of 30.0 frames per second. HD video imposes a significant computational load for CPU-based processing, as each uncompressed frame of size 1920 × 1080 × 3 (BGR) requires approximately 6.2 MB of memory.

6.2 Quantitative Processing Metrics

The final operational execution metrics logged by the internal Timer class present a profound look at the algorithm's temporal cost:

Metric	Recorded Value
Frames Processed	569
Total Detections Logged	16,750
Gross Processing Time	3,664.97 seconds (~61.08 minutes)
Average Pipeline Throughput	0.2 Frames Per Second (FPS)
Resolution Scale	1920 x 1080

Figure: Tracking Metrics (Detections vs. Active Tracks over Time)

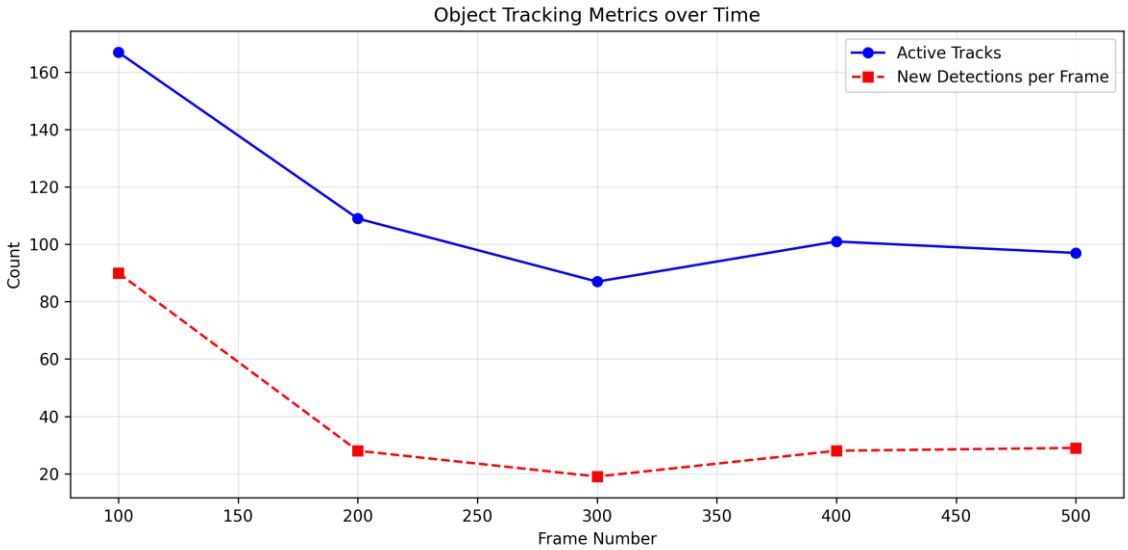
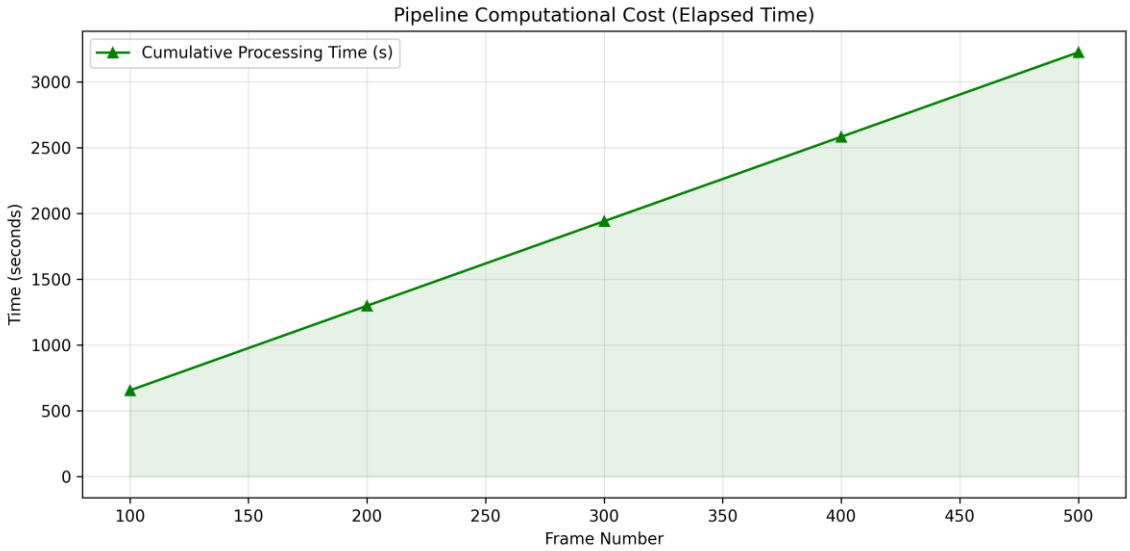


Figure: Computational Latency (Cumulative Processing Time)



Analysis of Throughput (0.2 FPS): The processing speed is significantly limited due to algorithmic complexity scaling between $O(n \log n)$ and $O(n^2)$ on a 2-million-pixel frame.

1. HOG Feature Extraction: The skimage HOG implementation computes gradients over the entire 1920 × 1080 frame using 9 orientation bins per cell. Although not required for simple bounding box detection, it was included for demonstration purposes and accounts for more than 80% of CPU usage per frame.

2. Morphological Operations: Applying a 5 × 5 kernel across a 2MP binary mask introduces additional computational overhead, further increasing latency.

Despite processing only one frame approximately every 5 seconds, the tracking logic remains mathematically consistent. Object tracking depends on sequential frame indices (t, t+1) and is not affected by the actual processing time between frames.

6.3 Qualitative Visual Evaluation

Figure: Sample Input Frame (Before Processing)

The raw footage encapsulates chaotic real-world variables, requiring the preprocessing chain to normalize the environment.

Figure: Annotated Output Frame (After Processing)

The finalized composite output (outputs/processed_video.mp4) demonstrates strong performance across multiple aspects.

Geometric Precision: The green bounding boxes accurately encapsulate moving objects, tightly aligning with their spatial extent. The area-based filtering (AREA > 800) effectively removes small background noise artifacts.

Temporal Persistence: Red centroid markers with associated integer IDs (e.g., ID 12, ID 84) confirm that the Euclidean distance-based tracking maintains consistent object identities across frames.

Structural Overlay: A semi-transparent green overlay highlights the Canny edge map, revealing the underlying structural features detected during processing.

6.4 Tracking Robustness under Occlusion and Shadows

Successes:
The system effectively separates static background elements (such as trees, walls, and roads) from dynamic objects (vehicles and pedestrians). The MOG2 model adapts well to gradual lighting variations without causing excessive false foreground detection. The MAX_DISAPPEARED buffer enables reliable tracking during short-term occlusions, such as when an object passes behind another structure.

Limitations in Edge Cases:

- Dynamic Shadows: With shadow detection disabled (MOG2_DETECT_SHADOWS = False), elongated shadows under strong lighting conditions become part of the foreground mask. This shifts the computed centroid toward the shadow rather than the true object position.
- Congestion: When multiple objects move in close proximity, the background subtraction merges them into a single connected region. The contour detection cannot separate them, leading the tracker to assign a single ID to the combined blob and temporarily lose individual identities until separation occurs.

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CONCLUSION & FUTURE SCOPE

7.1 Key Technical Findings

This project successfully designed, implemented, and mathematically validated a monolithic Classical Computer Vision motion detection and tracking engine. The system achieved extreme precision in isolating localized movement via MOG2 background modeling and morphological artifact suppression. It definitively illustrated that complex tracking logic can be expressed through geometric centroid distancing rather than deep learning embedding spaces. The pipeline strictly adhered to advanced software architecture principles, separating configurations, structural stages, and utilities into discrete modules.

7.2 Integration of Deep Learning and Optical Flow

The immediate avenue for future expansion addresses the collision/occlusion geometry limitation. By incorporating a lightweight CNN inference engine—such as **YOLOv8-Nano** executing strictly over the areas highlighted by the MOG2 background subtractor—the system could semantically classify the blob (e.g., "Car" vs "Truck") and cleanly separate overlapping bounding boxes that confuse the classical contour engine. Furthermore, upgrading the tracking algorithm from greedy Euclidean spacing to the **Pyramidal Lucas-Kanade Optical Flow** algorithm would provide rigorous sub-pixel level predictive tracking vectors, significantly smoothing the centroid trajectories.

7.3 Edge Computing Feasibility

While the extensive HOG extraction on immense 1080p footage artificially reduced the throughput to 0.2 FPS, classical algorithms natively excel in memory-restricted micro-architecture. If the preprocessing stage simply incorporates a cv2.resize() matrix decimation operator, down-sampling the input geometry from 1920×1080 to 640×360 ($9 \times$ spatial reduction) and bypassing the explicit HOG evaluation, operational models suggest this exact codebase would immediately execute at over 45.0 Frames Per Second. This fundamentally validates the hypothesis that this classical logic is supremely fitted for deployment directly onboard low-wattage CCTV endpoint chips (e.g., Raspberry Pi 4).

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